

API Reference

ETSI GS QKD 004 push mode specification, KMS server

API Version: 1.0.0

Server description of the KMS endpoint processing ETSI GS QKD 004 push mode requests

CONTACT

URL: <https://qkd-kms.ait.ac.at>

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SECURITY SCHEMES

KEY	TYPE	DESCRIPTION
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API

1. ETSI004

1.1 POST /key-material

endpoint to push key material to.

According to the sequence, after a successful get_key, the QKD can push the corresponding key material to this endpoint, whenever it created a new key.

REQUEST

REQUEST BODY - application/json

```
{
  status*      enum      ALLOWED:0, 1, 2, 3, 4, 5, 6, 7, 8
                                Status code:
                                0 = Successful
                                1 = Successful connection, but peer not connected
                                2 = GET_KEY failed because insufficient key available
                                3 = GET_KEY failed because peer application is not yet connected
                                4 = No QKD connection available
                                5 = OPEN_CONNECT failed because the KSID is already in use
                                6 = TIMEOUT_ERROR: The call failed because the specified TIMEOUT
                                7 = OPEN failed because requested QoS settings could not be met, counter proposal included in
                                8 = GET_KEY failed because metadata field size insufficient. Returned Metadata_size value holds
                                minimum needed size of metadata
                                increasing index of the key chunk in the key stream. Wrapping at UINT_MAX.

  index*       integer
  key_buffer*  string
  metadata {
    size        integer          size of buffer in characters
    buffer {
      metadata in json format.
    }
    key_stream_id* string key stream ID to which the value belongs (required for KMS in case of multiple open key
                                streams)
  }
}
```

RESPONSE

STATUS CODE - 201: Key successfully received

RESPONSE MODEL - application/json

```
{
  status*      enum      ALLOWED:0, 1, 2, 3, 4, 5, 6, 7, 8
                                Status code:
                                0 = Successful
                                1 = Successful connection, but peer not connected
                                2 = GET_KEY failed because insufficient key available
                                3 = GET_KEY failed because peer application is not yet connected
                                4 = No QKD connection available
                                5 = OPEN_CONNECT failed because the KSID is already in use
                                6 = TIMEOUT_ERROR: The call failed because the specified TIMEOUT
                                7 = OPEN failed because requested QoS settings could not be met, counter proposal included in return has
                                occurred
                                8 = GET_KEY failed because metadata field size insufficient. Returned Metadata_size value holds minimum
                                needed size of metadata

  metadata {
    size        integer          size of buffer in characters
    buffer      string          metadata in json format.
  }
}
```